

Paper Review

Faithlessness in Gaussian graphical models.

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· Drton, M., Henckel, L., Hollering, B., & Misra, P. (2024.)

Faithlessness in Gaussian graphical models.

arXiv preprint arXiv:2404.05306.

Main contributions and outline

Main contributions

- Provide an algebraic perspective for investigating the cases where faithfulness does not hold in (Gaussian) linear SEMs.
- Investigate propagation of faithlessness and strong faithlessness respectively.

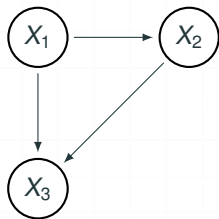
- Introduction
- Preliminaries
- The algebraic geometry of unfaithful distributions
- Decomposition into graphical models
- Violation of strong faithfulness
- Discussion

Introduction

Introduction

- It is important to understand, given a set of conditional independence statements which **other independence statements have to hold in any distribution that satisfies the former.**
- For general collections of conditional independence statements, it has been shown that **no finite axiomatization to solve the implication problem exists.**
- As a result, most research on probabilistic and causal reasoning has focused on probabilistic graphical models, where **the implication problem can be solved in polynomial time.**
- This paper studies the problems of **faithlessness and δ -faithlessness propagation** for linear (Gaussian) DAG models.

Motivating example



$\mathcal{G}$: Three-node DAG

- Consider a three-node linear SEMs:

$$X_1 = \epsilon_1, \quad X_2 = X_1 + \epsilon_2, \quad X_3 = X_1 + X_2 + \epsilon_3,$$

with $\epsilon_i \stackrel{i.i.d.}{\sim} N(0, 1)$.

- Its inverse covariance matrix is as follows:

$$\Sigma^{-1} = \begin{pmatrix} 3 & 0 & -1 \\ 0 & 2 & -1 \\ -1 & -1 & 1 \end{pmatrix}.$$

- There is an edge from X_1 to X_2 , but its partial covariance is 0, which implies $X_1 \perp\!\!\!\perp X_2 \mid X_3$.
- This is the case where the faithfulness assumption is violated.

Preliminaries

Preliminaries

- Let $\mathcal{G} = (V, E)$ be a directed acyclic graph (DAG) with node set V and edge set E , and let $\epsilon = (\epsilon_i \mid i \in V)$ be a (Gaussian) random vector with mean 0 and diagonal covariance matrix Ω .
- A random vector X is distributed according to the **linear structural equation model** on $\mathcal{G}$:

$$X_j = \sum_{i \in \text{Pa}(j)} \lambda_{ij} X_i + \epsilon_j,$$

where λ_{ij} are edge weights.

- Thus any random vector satisfying the above linear SEM will also be (Gaussian) with mean 0 and covariance matrix Σ given by

$$\Sigma = (I - \Lambda)^{-T} \Omega (I - \Lambda)^{-1},$$

where $\Lambda = (\lambda_{ij})$.

Definition 2.1. (Graphical model)

Let $\mathcal{G} = (V, E)$ be a DAG with n nodes. (Identifying centered Gaussian joint distributions with their covariance matrix,) we define the (Gaussian) linear structural equation model on $\mathcal{G}$ as the **image of the map**

$$\begin{aligned}\phi_{\mathcal{G}} : \mathbb{R}^E \times (0, \infty)^n &\rightarrow \text{PD}_n \\ (\Lambda, \Omega) &\mapsto (I - \Lambda)^{-T} \Omega (I - \Lambda)^{-1},\end{aligned}$$

where PD_n is the cone $n \times n$ positive definite matrices. We denote this model by $\mathcal{M}_{\mathcal{G}} = \text{image}(\phi_{\mathcal{G}})$. Subsequently, we will refer to $\mathcal{M}_{\mathcal{G}}$ simply as the (Gaussian) DAG model on $\mathcal{G}$.

- It is allowed for an edge weight to be zero.
- Intuitively, the graphical model is a set of possible covariance matrices under the given DAG $\mathcal{G}$.

Preliminaries

- $X_A \perp\!\!\!\perp X_B \mid X_C$ or $(A \perp\!\!\!\perp B \mid C)$: X_A and X_B are **conditionally independent** given X_C .
- $X_A \perp\!\!\!\perp_{\Sigma} X_B \mid X_C$ or $(A \perp\!\!\!\perp_{\Sigma} B \mid C)$: X_A and X_B are **partially uncorrelated** given X_C .
- $X_A \perp\!\!\!\perp_{\mathcal{G}} X_B \mid X_C$ or $(A \perp\!\!\!\perp_{\mathcal{G}} B \mid C)$: A and B are **d-separated** by C in $\mathcal{G}$.

Theorem 2.4. (D-separation and partial covariance)

A and B are d-separated by C in $\mathcal{G}$ if and only if the $X_A \perp\!\!\!\perp_{\Sigma} X_B \mid X_C$ holds for **every distribution in the graphical model associated to $\mathcal{G}$** .

- Note that the faithfulness assumes A and B are d-separated by C if $X_A \perp\!\!\!\perp_{\Sigma} X_B \mid X_C$.
- The faithfulness assumption would be violated if there exists $\tilde{\Sigma} \in \mathcal{M}_{\mathcal{G}}$ such that $X_A \not\perp\!\!\!\perp_{\tilde{\Sigma}} X_B \mid X_C$, which implies A and B are d-connected by C .
- It is known that violation of faithfulness only happens in **Lebesgue measure zero** set.

Preliminaries

- Let $C = \{I_1 \perp\!\!\!\perp J_1 \mid K_1, \dots, I_\ell \perp\!\!\!\perp J_\ell \mid K_\ell\}$ be a set of CI statements. Then, the (Gaussian) **conditional independence model** associated to C is the set of covariance matrices

$$\mathcal{M}_C = \{\Sigma \in \text{PD}_n : I \perp\!\!\!\perp J \mid K \text{ for all } I \perp\!\!\!\perp J \mid K \in C\}.$$

- A set of CI statements defined by the **global Markov property** would be

$$\text{global}(\mathcal{G}) = \{X_i \perp\!\!\!\perp X_j \mid X_k : i \perp\!\!\!\perp j \mid k\},$$

while that defined by the **local Markov property** is

$$\text{local}(\mathcal{G}) = \{X_i \perp\!\!\!\perp X_{\text{Nd}(i) \setminus \text{Pa}(i)} \mid X_{\text{Pa}(i)} : i \in V\}.$$

- By theorem 2.4 and Hammersley-Clifford theorem, we obtain

$$\mathcal{M}_{\mathcal{G}} = \mathcal{M}_{\text{global}(\mathcal{G})} = \mathcal{M}_{\text{local}(\mathcal{G})}.$$

Lemma. (CI statement and partial covariance)

$X_i \perp\!\!\!\perp_{\Sigma} X_j \mid X_K$ holds in a random vector $X \sim (\mu, \Sigma)$ if and only if $\det(\Sigma_{iK,jK}) = 0$, where

$$\Sigma_{iK,jK} = \begin{pmatrix} \sigma_{ij} & \Sigma_{iK} \\ \Sigma_{Kj} & \Sigma_{KK} \end{pmatrix}.$$

Proof.

$$\det(\Sigma_{iK,jK}) = (\sigma_{ij} - \Sigma_{iK} \Sigma_{KK}^{-1} \Sigma_{Kj}) \det(\Sigma_{KK}) = 0$$

$$\iff \sigma_{ij} - \Sigma_{iK} \Sigma_{KK}^{-1} \Sigma_{Kj} = 0$$

$$\iff X_i \perp\!\!\!\perp_{\Sigma} X_j \mid X_K. \quad \square$$

The algebraic geometry of unfaithful distributions

Problem 3.2.

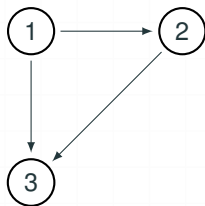
Let $\mathcal{M}_{\mathcal{G}}$ be the graphical model on the DAG $\mathcal{G}$ and suppose $i \not\perp_{\mathcal{G}} j \mid K$. For which disjoint sets $A, B, C \subset [n]$ does $\Sigma \in \mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{i \perp j \mid K}$ imply $X_A \perp_{\Sigma} X_B \mid X_C$?

- For $\Sigma \in \mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{i \perp j \mid K}$, $X_i \perp_{\Sigma} X_j \mid X_K$ even if i and j are d-connected by K , which is the case where faithfulness does not hold.
- Problem 3.2. aims to investigate a condition for Σ to be unfaithful with respect to a triple (X_i, X_j, X_K) .
- In some cases, this condition would unintentionally involve additional CI statements.

Decomposition of covariance

Definition 3.3. (Trek)

A **trek** in $\mathcal{G}$ from a node i to a node j is a pair (P_L, P_R) , where P_L is a directed path from some node s to i and P_R is a directed path from the same node s to j . Here, s is called the **topmost** node of the trek, whereas i and j are called the **leftmost** and **rightmost** vertices of the trek, respectively.



$\mathcal{G}$: Three-node DAG

- $2 \leftarrow 1 \rightarrow 3$
 - Topmost node : 1.
- $2 \leftarrow 1 \rightarrow 2 \rightarrow 3$
 - Topmost node : 1.
- $2 \leftarrow 2 \rightarrow 3$
 - Topmost node : 2.

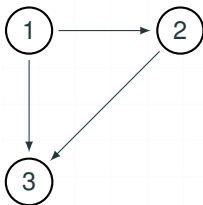
Decomposition of covariance

Definition 3.3. (Trek monomial)

For any given trek $T = (P_L, P_R)$, the associated **trek monomial** m_T is given by

$$m_T = w_s \prod_{k \rightarrow \ell \in P_L} \lambda_{k\ell} \prod_{k \rightarrow \ell \in P_R} \lambda_{k\ell}.$$

- With this terminology in place, we may formulate the **trek rule** which states that the entries of any covariance matrix $\Sigma \in \mathcal{M}_{\mathcal{G}}$ is given by $\sigma_{ij} = \sum_T m_T$.



$\mathcal{G}$: Three-node DAG

- $2 \leftarrow 1 \rightarrow 3$
 - $\triangleright m_T = \omega_1 \lambda_{12} \lambda_{13}.$
- $2 \leftarrow 1 \rightarrow 2 \rightarrow 3$
 - $\triangleright m_T = \omega_1 \lambda_{12}^2 \lambda_{23}.$
- $2 \leftarrow 2 \rightarrow 3$
 - $\triangleright m_T = \omega_2 \lambda_{23}.$

Decomposition of covariance

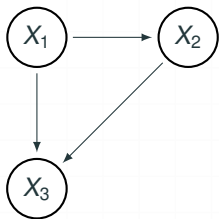
Definition 3.3. (Trek rule)

Let $\mathbb{R}[\Sigma] = \mathbb{R}[\sigma_{ij} \mid 1 \leq i \leq j \leq n]$ and $\mathbb{R}[\lambda, \omega]$ be the rings of polynomials in the covariances σ_{ij} and the non-zero entries of (Λ, Ω) , respectively. Then algebraically, the **trek rule** can be seen as a ring homomorphism $\phi_{\mathcal{G}}^*$ from $\mathbb{R}[\Sigma]$ to $\mathbb{R}[\lambda, \omega]$. Specifically,

$$\begin{aligned}\phi_{\mathcal{G}}^* : \mathbb{R}[\Sigma] &\rightarrow \mathbb{R}[\lambda, \omega] \\ \sigma_{ij} &\mapsto \sum_T m_T.\end{aligned}$$

- $\phi_{\mathcal{G}}^*(f)$ compactly denotes the result of substituting the trek rule expressions for each of the covariances appearing in the polynomial $f \in \mathbb{R}[\Sigma]$.
- Continuing the example, $\phi_{\mathcal{G}}^*(\sigma_{23}) = \omega_1 \lambda_{12} \lambda_{13} + \omega_1 \lambda_{12}^2 \lambda_{23} + \omega_2 \lambda_{23}$.

Characterization of faithfulness



$\mathcal{G}$: Three-node DAG

- $\sigma_{12} = \omega_1 \lambda_{12}$.
- $\sigma_{33} = \omega_1 (\lambda_{13} + \lambda_{12} \lambda_{23})^2 + \omega_2 \lambda_{23}^2 + \omega_3$.
- $\sigma_{13} = \omega_1 (\lambda_{13} + \lambda_{12} \lambda_{23})$.
- $\sigma_{23} = \omega_1 \lambda_{12} (\lambda_{13} + \lambda_{12} \lambda_{23}) + \omega_2 \lambda_{23}$.

- Our goal is to find a parameter space where the distribution of X is unfaithful with respect to $X_1 \not\perp_{\mathcal{G}} X_2 \mid X_3$.
- By the previous lemma, $\det(\Sigma_{13,23})$ should be zero, which can be expressed by (Λ, Ω) using the trek rule.

$$\begin{aligned}\phi_{\mathcal{G}}^*(\det(\Sigma_{13,23})) &= \phi_{\mathcal{G}}^*(\sigma_{12}\sigma_{33} - \sigma_{13}\sigma_{23}) \\ &= \omega_1(\omega_3\lambda_{12} - \omega_2\lambda_{13}\lambda_{23}) \\ &= 0.\end{aligned}$$

- Finally, we obtain

$$\begin{aligned}\mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{1 \not\perp 2 \mid 3} \\ = \phi_{\mathcal{G}}\left(\{(\Lambda, \Omega) \in \mathbb{R}^E \times (0, \infty)^n : \omega_3\lambda_{12} - \omega_2\lambda_{13}\lambda_{23} = 0\}\right).\end{aligned}$$

Characterization of faithlessness

Theorem 3.4. (Characterization of faithlessness)

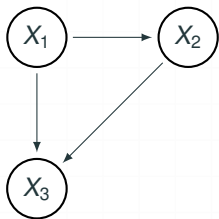
Let $\mathcal{G} = (V, E)$ be a DAG and $i \perp\!\!\!\perp j \mid K$ be a CI statement which is not implied by d-separation on $\mathcal{G}$, i.e. i and j are d-connected by K . Let $\phi_{\mathcal{G}}^*$ be the ring map which corresponds to the parameterization of $\mathcal{M}_{\mathcal{G}}$ and $\phi_{\mathcal{G}}^*(\det(\Sigma_{iK,jK})) = \prod_{\ell=1}^m f_{\ell}$ be an irreducible factorization. Then

$$\mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j \mid K} := \mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{i \perp\!\!\!\perp j \mid K} = \bigcup_{\ell=1}^m \phi_{\mathcal{G}}(\mathcal{V}(f_{\ell})) \cap \text{PD}_n,$$

where $\mathcal{V}(f_{\ell}) = \{(\Lambda, \Omega) \in \mathbb{R}^E \times (0, \infty)^n : f_{\ell} = 0\}$, and $\phi_{\mathcal{G}}(\mathcal{V}(f_{\ell})) = \{\phi_{\mathcal{G}}(\Lambda, \Omega) : (\Lambda, \Omega) \in \mathcal{V}(f_{\ell})\}$.

- Each element of $\mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j \mid K}$ is a covariance matrix that satisfies all d-separation statements in $\mathcal{G}$ while X_i and X_j are partially uncorrelated given X_K .
- Recall that $X_i \perp\!\!\!\perp_{\Sigma} X_j \mid X_K$ holds if and only if $\det(\Sigma_{iK,jK}) = 0$.
- This requires at least one of f_{ℓ} to be zero, where f_{ℓ} is a function of (Λ, Ω) according to the definition of $\phi_{\mathcal{G}}^*$.

Characterization of faithfulness



$\mathcal{G}$: Three-node DAG

- $\sigma_{12} = \omega_1 \lambda_{12}$.
- $\sigma_{33} = \omega_1 (\lambda_{13} + \lambda_{12} \lambda_{23})^2 + \omega_2 \lambda_{23}^2 + \omega_3$.
- $\sigma_{13} = \omega_1 (\lambda_{13} + \lambda_{12} \lambda_{23})$.
- $\sigma_{23} = \omega_1 \lambda_{12} (\lambda_{13} + \lambda_{12} \lambda_{23}) + \omega_2 \lambda_{23}$.

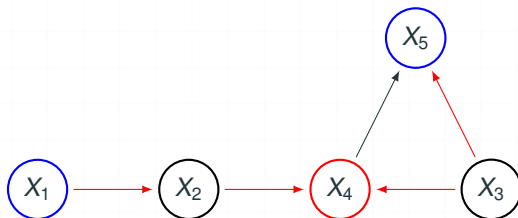
- Our goal is to find a parameter space where the distribution of X is unfaithful with respect to $X_1 \not\perp_{\mathcal{G}} X_2 \mid X_3$.
- By the previous lemma, $\det(\Sigma_{13,23})$ should be zero, which can be expressed by (Λ, Ω) using the trek rule.

$$\begin{aligned}\phi_{\mathcal{G}}^*(\det(\Sigma_{13,23})) &= \phi_{\mathcal{G}}^*(\sigma_{12}\sigma_{33} - \sigma_{13}\sigma_{23}) \\ &= \omega_1(\omega_3\lambda_{12} - \omega_2\lambda_{13}\lambda_{23}) \\ &= 0.\end{aligned}$$

- Finally, we obtain

$$\begin{aligned}\mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{1 \perp\!\!\!\perp 2 \mid 3} \\ = \phi_{\mathcal{G}}\left(\{(\Lambda, \Omega) \in \mathbb{R}^E \times (0, \infty)^n : \omega_3\lambda_{12} - \omega_2\lambda_{13}\lambda_{23} = 0\}\right).\end{aligned}$$

Characterization of faithfulness



$\mathcal{G}$: Five-node DAG

- Note that $X_1 \not\perp_{\mathcal{G}} X_5 \mid X_4$.
- Using the trek rule, one would obtain $\phi_{\mathcal{G}}^*(\det(\Sigma_{14,45})) = \omega_1 \omega_3 \lambda_{12} \lambda_{24} \lambda_{34} \lambda_{35}$.
- Then, $\mathcal{M}_{\mathcal{G}, 1 \perp\!\!\!\perp 5 \mid 4} = \mathcal{M}_{\mathcal{G} \setminus 1 \rightarrow 2} \cup \mathcal{M}_{\mathcal{G} \setminus 2 \rightarrow 4} \cup \mathcal{M}_{\mathcal{G} \setminus 3 \rightarrow 4} \cup \mathcal{M}_{\mathcal{G} \setminus 3 \rightarrow 5}$, where $\mathcal{M}_{\mathcal{G} \setminus i \rightarrow j} = \phi_{\mathcal{G}}(\{(\Lambda, \Omega) \in \mathbb{R}^E \times (0, \infty)^n : \lambda_{ij} = 0\})$.
- Intuitively, $\mathcal{M}_{\mathcal{G} \setminus i \rightarrow j}$ corresponds to a sub-model of $\mathcal{M}_{\mathcal{G}}$ where the edge $i \rightarrow j$ is deleted.

Propagation of faithlessness - General

Corollary 3.6. (Propagation of faithlessness - General)

Let $a, b \in [n]$ be disjoint elements and $C \subseteq [n] \setminus \{a, b\}$. Then $a \perp\!\!\!\perp_{\Sigma} b \mid C$ for all distributions $\Sigma \in \mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j \mid K}$ if

$$\phi_{\mathcal{G}}^*(\det(\Sigma_{aC, bC})) \in \left\langle \phi_{\mathcal{G}}^*(\det(\Sigma_{iK, jK})) \right\rangle : \prod_{S \subseteq V} \phi_{\mathcal{G}}^*(\det(\Sigma_{S, S}))^{\infty}.$$

- Corollary 3.6 provides a sufficient condition for testing additional CI implications in $\mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j \mid K}$.
- Let $\phi_{\mathcal{G}}^*(\det(\Sigma_{iK, jK})) = \prod_{\ell=1}^m f_{\ell}$ be an irreducible factorization, and let $I = \left\{ \ell \in [m] : \exists S \subseteq V, \exists q \geq 1 \text{ s.t. } \phi_{\mathcal{G}}^*(\det(\Sigma_{S, S})) \propto f_{\ell}^{1/q} \right\}$.
- Considering that $\Sigma_{S, S}$ is a marginal covariance matrix w.r.t. X_S , its determinant is clearly non-zero, which implies that $f_{\ell} \neq 0 \quad \forall \ell \in I$.
- $\left\langle \phi_{\mathcal{G}}^*(\det(\Sigma_{iK, jK})) \right\rangle : \prod_{S \subseteq V} \phi_{\mathcal{G}}^*(\det(\Sigma_{S, S}))^{\infty} = \left\langle \prod_{\ell \notin I} f_{\ell} \right\rangle$.
- Intuitively, $\left\langle \phi_{\mathcal{G}}^*(\det(\Sigma_{iK, jK})) \right\rangle : \prod_{S \subseteq V} \phi_{\mathcal{G}}^*(\det(\Sigma_{S, S}))^{\infty}$ **only consists of factors that can be zero-valued.**

Propagation of faithlessness - General

Corollary 3.6. (Propagation of faithlessness - General)

Let $a, b \in [n]$ be disjoint elements and $C \subseteq [n] \setminus \{a, b\}$. Then $a \perp\!\!\!\perp_{\Sigma} b \mid C$ for all distributions $\Sigma \in \mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j \mid K}$ if

$$\phi_{\mathcal{G}}^*(\det(\Sigma_{aC, bC})) \in \left\langle \phi_{\mathcal{G}}^*(\det(\Sigma_{iK, jK})) \right\rangle : \prod_{S \subseteq V} \phi_{\mathcal{G}}^*(\det(\Sigma_{S, S}))^{\infty}.$$

- $\left\langle \phi_{\mathcal{G}}^*(\det(\Sigma_{iK, jK})) \right\rangle : \prod_{S \subseteq V} \phi_{\mathcal{G}}^*(\det(\Sigma_{S, S}))^{\infty} = \left\langle \prod_{\ell \notin I} f_{\ell} \right\rangle$.
- Intuitively, $\left\langle \phi_{\mathcal{G}}^*(\det(\Sigma_{iK, jK})) \right\rangle : \prod_{S \subseteq V} \phi_{\mathcal{G}}^*(\det(\Sigma_{S, S}))^{\infty}$ **only consists of factors that can be zero-valued**.
- The sufficient condition in Corollary 3.6 asks if $\phi_{\mathcal{G}}^*(\det(\Sigma_{aC, bC}))$ is divided by $\prod_{\ell \notin I} f_{\ell}$.

Remark.

If $\phi_{\mathcal{G}}^*(\det(\Sigma_{iK, jK})) = 0$ implies $\phi_{\mathcal{G}}^*(\det(\Sigma_{aC, bC})) = 0$, $a \perp\!\!\!\perp_{\Sigma} b \mid C$ holds for all distributions $\Sigma \in \mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j \mid K}$.

Propagation of faithlessness - General

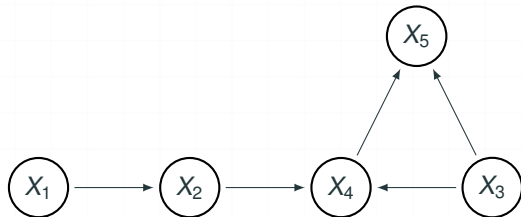
- Corollary 3.6 does still require to compute $\phi_{\mathcal{G}}^*(\det(\Sigma_{aC,bC}))$ and $\phi_{\mathcal{G}}^*(\det(\Sigma_{iK,jK}))$ which becomes much more difficult as the size of $\mathcal{G}$ grows.
- The corollary only provides a sufficient condition for testing CI implications.
- In general $\mathcal{M}_{\mathcal{G},i \perp\!\!\!\perp j|K}$ is no longer a graphical model, and hence finding CI statements using d-separation relationships is not available.
- By narrowing down to the cases where $\mathcal{M}_{\mathcal{G},i \perp\!\!\!\perp j|K}$ **can be expressed as a union of graphical models**, the CI implication problem can also be easily solved with d-separation.

Decomposition into graphical models

Propagation of faithlessness - Union of graphical models

Proposition 4.1. (Propagation of faithlessness - Union of graphical models)

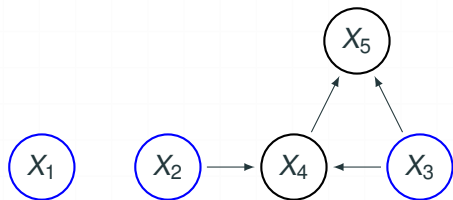
Let $\mathcal{G} = (V, E)$ be a DAG and $i \perp\!\!\!\perp j \mid K$ be a CI statement which is not implied by d-separation on $\mathcal{G}$. If $\mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j \mid K} = \bigcup_{\ell=1}^m \mathcal{M}_{\mathcal{G}_\ell}$ for some DAGs $\mathcal{G}_1, \dots, \mathcal{G}_m$ then $A \perp\!\!\!\perp_\Sigma B \mid C$ for all $\Sigma \in \mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j \mid K}$ if and only if $A \perp\!\!\!\perp_{\mathcal{G}_\ell} B \mid C$ for all $\ell \in [m]$.



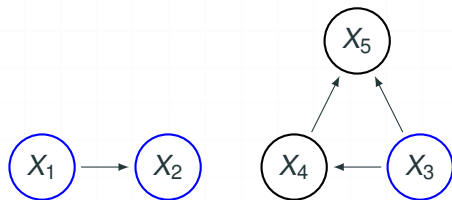
$\mathcal{G}$: Five-node DAG

- Recall that $\mathcal{M}_{\mathcal{G}, 1 \perp\!\!\!\perp 5 \mid 4} = \mathcal{M}_{\mathcal{G} \setminus 1 \rightarrow 2} \cup \mathcal{M}_{\mathcal{G} \setminus 2 \rightarrow 4} \cup \mathcal{M}_{\mathcal{G} \setminus 3 \rightarrow 4} \cup \mathcal{M}_{\mathcal{G} \setminus 3 \rightarrow 5}$.
- $X_1 \perp\!\!\!\perp_{\mathcal{G}_\ell} X_3$ and $X_2 \perp\!\!\!\perp_{\mathcal{G}_\ell} X_3$ for all $\ell = 1, 2, 3, 4$.
- This implies that $X_1 \perp\!\!\!\perp_\Sigma X_3$ and $X_2 \perp\!\!\!\perp_\Sigma X_3$ for all $\Sigma \in \mathcal{M}_{\mathcal{G}, 1 \perp\!\!\!\perp 5 \mid 4}$.

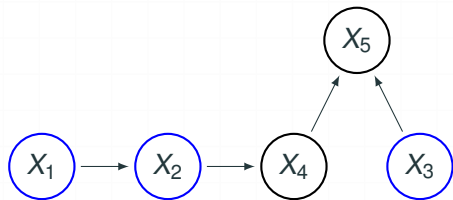
Propagation of faithlessness - Union of graphical models



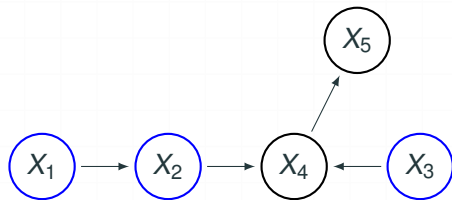
$\mathcal{G}_1 : \mathcal{G} \setminus 1 \rightarrow 2$



$\mathcal{G}_2 : \mathcal{G} \setminus 2 \rightarrow 4$



$\mathcal{G}_3 : \mathcal{G} \setminus 3 \rightarrow 4$



$\mathcal{G}_4 : \mathcal{G} \setminus 3 \rightarrow 5$

When $\mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j | K}$ is a union of graphical models

Theorem 4.8. (Necessary and sufficient condition for $\mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j | K}$ to be a union of graphical models)

Let $\mathcal{G} = (V, E)$ be a DAG and $\{i\}, \{j\}, K \subset V$ be sets of vertices such that K does not d-separate i from j . Then, there exist graphs $\mathcal{G}_1, \dots, \mathcal{G}_m$ such that $\mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j | K} = \bigcup_{\ell=1}^m \mathcal{M}_{\mathcal{G}_\ell}$ if and only if

- there is a unique d-connecting path from i to j given K (denoted by $i \leftrightarrow j$), and
 - for all $k_s \notin i \leftrightarrow j$, it holds that $\text{Pa}(k_s) \subseteq i \leftrightarrow j \cup K$.
-
- Theorem 4.8 provides a necessary and sufficient condition for $\mathcal{M}_{\mathcal{G}, i \perp\!\!\!\perp j | K}$ to be a union of graphical models.
 - However, the exact forms of $\mathcal{G}_1, \dots, \mathcal{G}_m$ are still required to apply Proposition 4.1 to detect a propagation.

Violation of strong faithfulness

Definition. (δ -faithfulness)

Let $\mathcal{G} = (V, E)$ a DAG and $\mathcal{M}_{\mathcal{G}}$ be an associated graphical model. We say $\mathcal{G}$ is δ -faithful for $\delta > 0$ if

$$i \perp\!\!\!\perp_{\mathcal{G}} j \mid K \iff |\rho_{i,j \cdot K}| \leq \delta,$$

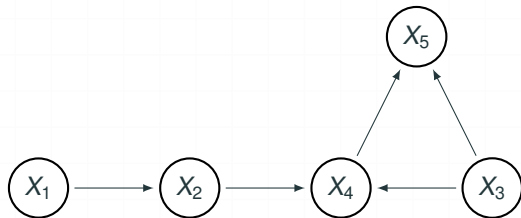
where $\rho_{i,j \cdot K}$ denotes a partial correlation coefficient between X_i and X_j given X_K .

- The notion of faithfulness has been criticized since near conditional independence cannot be differentiated from exact conditional independence in finite sample.
- In response the notion of δ -faithfulness has been proposed as an alternative.
- This raises a natural question: How does propagation of δ -faithfulness occur?

Propagation of δ -faithfulness

Theorem 6.1. (Propagation of δ -faithfulness)

Let $\mathcal{G} = (V, E)$ be a DAG and $\{i\}, \{j\}, K \subset V$ be sets of vertices such that K does not d-separate i from j . Let $\mathcal{M}_{i \perp_{\delta} j | K}$ denote the set of covariance matrices such that $|\rho_{i,j,K}| \leq \delta$, and consider a node ℓ such that K does not d-separate i from ℓ . Then for all $\delta \in [0, 1]$, $\mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{i \perp_{\delta} j | K} \subseteq \mathcal{M}_{i \perp_{\delta} \ell | K}$ if and only if $i \perp_{\mathcal{G}} \ell \mid \{j\} \cup K$.

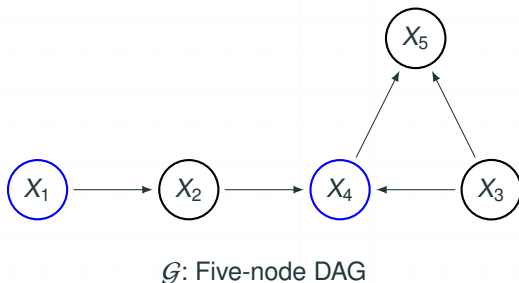


$\mathcal{G}$: Five-node DAG

- Note that
 - $1 \not\perp_{\mathcal{G}} 2 \mid 3$ and $1 \perp_{\mathcal{G}} 4 \mid 2, 3$.
 - $1 \not\perp_{\mathcal{G}} 4 \mid 3$ and $1 \perp_{\mathcal{G}} 5 \mid 4, 3$.
- Then propagation appears as follows.

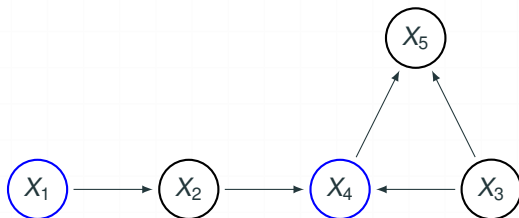
$$\begin{aligned}\mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{1 \perp_{\delta} 2 | 3} &= \mathcal{M}_{\mathcal{G}} \cap (\mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{1 \perp_{\delta} 2 | 3}) \\ &\subseteq \mathcal{M}_{\mathcal{G}} \cap \mathcal{M}_{1 \perp_{\delta} 4 | 3} \\ &\subseteq \mathcal{M}_{1 \perp_{\delta} 5 | 3}.\end{aligned}$$

Comparison between faithfulness and δ -faithfulness



- Simple algebra gives $\mathcal{M}_{\mathcal{G}, 1 \perp\!\!\!\perp 4} = \mathcal{M}_{\mathcal{G} \setminus 1 \rightarrow 2} \cup \mathcal{M}_{\mathcal{G} \setminus 2 \rightarrow 4}$.
- Since it is clear that 1 and 5 are d-separated in both sub-graphs, $X_1 \perp\!\!\!\perp_{\Sigma} X_5$ holds for all $\Sigma \in \mathcal{M}_{\mathcal{G}, 1 \perp\!\!\!\perp 4}$ by Proposition 4.1.

Comparison between faithfulness and δ -faithfulness



$\mathcal{G}$: Five-node DAG

- However, 1 and 5 are d-connected by 4, i.e. $1 \not\perp_{\mathcal{G}} 5 \mid 4$.
- According to Theorem 6.1, $X_1 \perp_{\delta} X_4$ does not always imply $X_1 \perp_{\delta} X_5$.
- This example illustrates that **the propagation of faithfulness and δ -faithfulness are distinct**.

Discussion

Discussion

- This paper studies the problems of faithfulness and δ -faithfulness propagation for linear (Gaussian) DAG models.
- It offers a necessary and sufficient condition for the propagation of faithlessness when the unfaithful model can be decomposed into a union of graphical models.
- However, it still requires the exact structure of each sub-graphical model, which can be attained by calculating its determinant.
- Additionally, this framework cannot be applied under non-zero edge weight condition.
- There is no structural constraint to detect a propagation of δ -faithlessness, but it involves an iterative process which would be infeasible in a large graph.